Statistics & Probability

Section 1: Measures of Central Tendency & Dispersion

1 The numbers $4, 7, 9, 12, 18$ have a mean of $\bar{x} = 10$. Find the standard deviation σ correct to 2 decimal places.

a. 4.73

b. 5.21

c. 4.12

d. 3.85

2 A data set of 20 observations has $\sum x = 240$ and $\sum x^2 = 3280$. Find the variance of the data set.

a. 20

b. 16

c. 24

d. 12

3 In a box-and-whisker plot, the lower quartile is $Q_1 = 24$ and the upper quartile is $Q_3 = 42$. According to the $1.5 \times \text{IQR}$ rule, what is the boundary for outliers at the upper end?

a. 69

b. 60

c. 65

d. 72

Section 2: Probability Rules & Conditional Probability

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4 Two independent events A and B have $P(A) = 0.4$ and $P(B) = 0.5$. Find $P(A \cup B)$.

a. 0.70

b. 0.90

c. 0.20

d. 0.65

5 Given $P(A) = 0.6$, $P(B) = 0.5$, and $P(A \cap B) = 0.3$, find the conditional probability $P(A \mid B)$.

a. 0.60

b. 0.50

c. 0.30

d. 0.75

6 A bag contains 4 red balls and 6 blue balls. Two balls are drawn at random without replacement. What is the probability that both balls are red?

a. $\frac{2}{15}$

b. $\frac{4}{25}$

c. $\frac{1}{6}$

d. $\frac{1}{5}$

Section 3: Discrete Probability Distributions

7 A discrete random variable X has probability function $P(X = x) = kx$ for $x \in \{1, 2, 3, 4\}$. Find the value of k.

a. $\frac{1}{10}$

b. $\frac{1}{6}$

c. $\frac{1}{4}$

d. $\frac{1}{12}$

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8 Find the expectation $E(X)$ for the distribution where $P(X=1)=0.2, P(X=2)=0.5, P(X=3)=0.3$.

a. 2.1

b. 2.0

c. 2.3

d. 1.9

9 For a discrete random variable X with $E(X)=4$ and $E(X^2)=25$, what is the variance $\text{Var}(X)$?

a. 9

b. 5

c. 16

d. 21

Section 4: Binomial Distribution & Exam Applications

10 A random variable $X \sim B(8, 0.25)$. Find $P(X=2)$ correct to 4 decimal places.

a. 0.3115

b. 0.2076

c. 0.2670

d. 0.1038

11 If $X \sim B(100, 0.4)$, find the mean and variance of X.

a. Mean $=40$, Variance $=24$

b. Mean $=40$, Variance $=16$

c. Mean $=50$, Variance $=25$

d. Mean $=40$, Variance $=4.9$

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12 A multiple-choice quiz has 5 questions, each with 4 options and exactly one correct answer. If a student guesses at random on all 5 questions, what is the probability of passing with at least 3 correct answers?

a. $\frac{53}{512} \approx 0.1035$

b. 0.2500

c. 0.0879

d. 0.1563

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